

R-coin: A Gold-Backed Global Currency for Universal Basic Income

Embedding redistribution in the monetary system through a union of central banks

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Contents

Executive summary	3
Introduction: from diagnosis to design	5
The economic foundation	5
Growth, stability, and the private-sector production rate	5
The equilibrium point P	7
The erosion of control in a multipolar world	8
The case for an autonomous global public sector	8
Financial policy: redistribution as the stabilising instrument	9
Technological deflation and the limits of conventional monetary policy	9
A formal model of redistribution	9
Universal basic income as the limiting case	11
Monetary policy: a gold-backed global currency	12
Revaluation: re-anchoring trust in gold, digitally	12
Transfer fees: redistribution without taxation	13
The dual mandate of the issuer	13
The R-coin concept	14
Redistribution, Revaluation, Revolution	14
Why not a decentralised cryptocurrency	14
The Central Bank Union	14
System architecture	15
The blockchain layer	15
Validators and consensus	15
Privacy, encryption, and the two-tier model	16
Access authorisation	16
Resilience and circulation control	17
A note on next-generation hardware	17
The universal basic income program	17
Sizing the grant	17
Funding mechanics	18
Inclusion: biometric identity and the digital divide	18
Universal basic income as a smart contract	19
Governance and operations of the CBU	19
Implementation roadmap	20
Risks, safeguards, and open questions	20
Governance lineage: the world federal government	21
Conclusion	22
Addendum	22
Appendix A — The Montreux Declaration: six principles	23
Appendix B — Mathematical summary of the redistribution model	23
Appendix C — Glossary	24
Appendix D — Sources and further reading	25

Executive summary

This white paper sets out a unified case for a single, gold-backed global currency whose defining feature is a redistribution mechanism embedded directly in the monetary system. It draws together three strands of work developed by the author since 2020: a macroeconomic argument for structural reform of global economic governance; a system design for a digital currency issued and operated by a union of central banks; and a concrete parameterisation of a universal basic income (UBI) program funded not by taxation but by transaction transfer fees.

The argument proceeds in three movements. First, it diagnoses a structural tension at the heart of the modern economy: the same private-sector dynamism that drives growth also amplifies the multiplier effects that destabilise it. The balance between growth and stability is set, in effect, by public-sector decisions — legislation, fiscal policy, and monetary policy. In a multipolar world, those decisions are fragmented across competing jurisdictions, and the control they once exerted has weakened. Persistent, technology-driven deflation and the side effects of unconventional monetary policy have widened wealth and income gaps to the point where they threaten social, and therefore economic, stability.

Second, the paper proposes redistribution as the stabilising instrument the system now lacks, and shows that a universal basic income is its simplest and most consensual form. Because more than half of the world's population would receive more than they pay in, such a program can command durable democratic support. The paper formalises the redistribution rule and derives the size of a globally meaningful grant.

Third, it describes how this redistribution can be implemented as a property of money itself. A blockchain-based global currency, with its unit anchored to the weight of gold, can collect a small transfer fee on the millions of transactions that occur every hour and recycle those fees into a basic income paid to every person — including those without a bank account, a smartphone, or internet access. The currency would be issued and governed by a Central Bank Union (CBU): an institution formed by the world's central banks, structurally independent in the manner of a judiciary, yet remaining within the public sector and accountable to the people who use the currency.

A worked illustration sets the grant at five units per person per year, where one unit equals one gram of gold. For the roughly 6.1 billion people aged fifteen and over, this implies an annual budget on the order of thirty thousand five hundred tonnes of gold-equivalent — issued once as a gold-denominated asset and thereafter sustained by transfer fees expected to remain below one percent. The paper closes with an implementation roadmap, a candid assessment of risks, and a note on the longer governance tradition — the world-federalist movement and its 1947 Montreux Declaration — within which this proposal sits.

The proposal is offered to central banks, finance ministries, and international institutions as a framework for discussion rather than a finished blueprint. Its purpose is to show

that monetary and social policy need not be separate: that the architecture of digital money can be designed, from the outset, to share the gains of technological progress.

Introduction: from diagnosis to design

For most of the modern era, the question of how the gains from economic growth should be shared has been treated as separate from the question of how money is issued and managed. Fiscal authorities tax and transfer; monetary authorities target inflation and safeguard the financial system. The two domains interact, but they are designed and operated apart.

This paper argues that the separation is becoming untenable, and that the arrival of programmable digital money makes a different arrangement possible for the first time. If a currency can carry a rule, then redistribution can be a property of the monetary system rather than an act of government layered on top of it. The remainder of this document develops that idea from first principles and then specifies a system capable of realising it.

The work synthesises and extends three earlier documents by the author. The macroeconomic foundation is taken from *The Global Economic Outlook* (2020), which argued for structural reform of global economic governance and introduced the redistribution and gold-backed-currency concepts.¹ The system design is taken from the R-coin litepaper (2022) and the 2023 presentation *Universal Basic Income as a Smart Contract*, which together specify the three-phase rollout, the validator and governance model, the data-access architecture, and the program's quantitative parameters.² Where the present paper introduces figures, corrected arithmetic, or additional policy framing, it does so to make the combined argument internally consistent and suitable for an institutional readership.

A note on naming. The working name *R-coin* stands for the three principles examined in this paper — **R**edistribution, **R**evaluation, and **R**evolution. The name is provisional: the intention is that the world community would rename the currency, through the system's own voting function, once it is established.

The economic foundation

Growth, stability, and the private-sector production rate

Consider the simplifying premise that all goods and services in the world are produced by two sectors, public and private, where *production* is understood broadly to include capital gains as well as conventional output. The public sector tends to supply goods

¹ Ryoji Furui, *The Global Economic Outlook — for the steady economic growth of the world*, 23 January 2020. Available at the project repository, <https://github.com/r-coin>.

² The R-coin project documents are maintained at <https://github.com/r-coin> and <https://r-coin.net>.

and services that are socially necessary but not highly profitable; the private sector tends to concentrate where profitability is higher.

The first relationship to observe is between the share of production carried by the private sector and the economy's rate of growth. Where more output is produced competitively by private actors, market competition rewards productivity, and the growth rate rises. The relationship is, to a first approximation, positive (Figure 1).

Economic growth rate vs. private-sector production

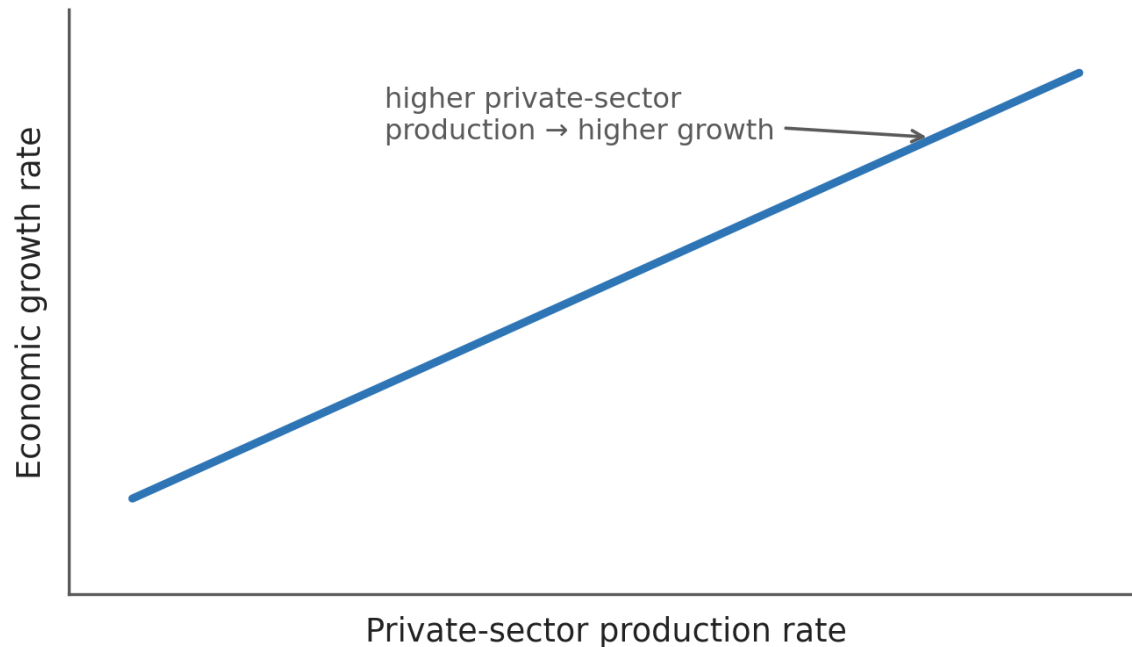


Figure 1. *As the private-sector share of production rises, competitive pressure favours productivity and the economy's growth rate increases.*

The second relationship runs in the opposite direction. The market mechanism that delivers equilibrium also transmits multiplier effects, and these can diverge as easily as they converge. Because the private sector contains far more independent decision-makers than the public sector, it generates more of these multiplier effects. Beyond a certain point, an economy dominated by private production becomes volatile and difficult to govern, as the dispersion of economic indicators widens. Economic stability therefore declines as the private-sector production rate rises (Figure 2).

Economic stability vs. private-sector production

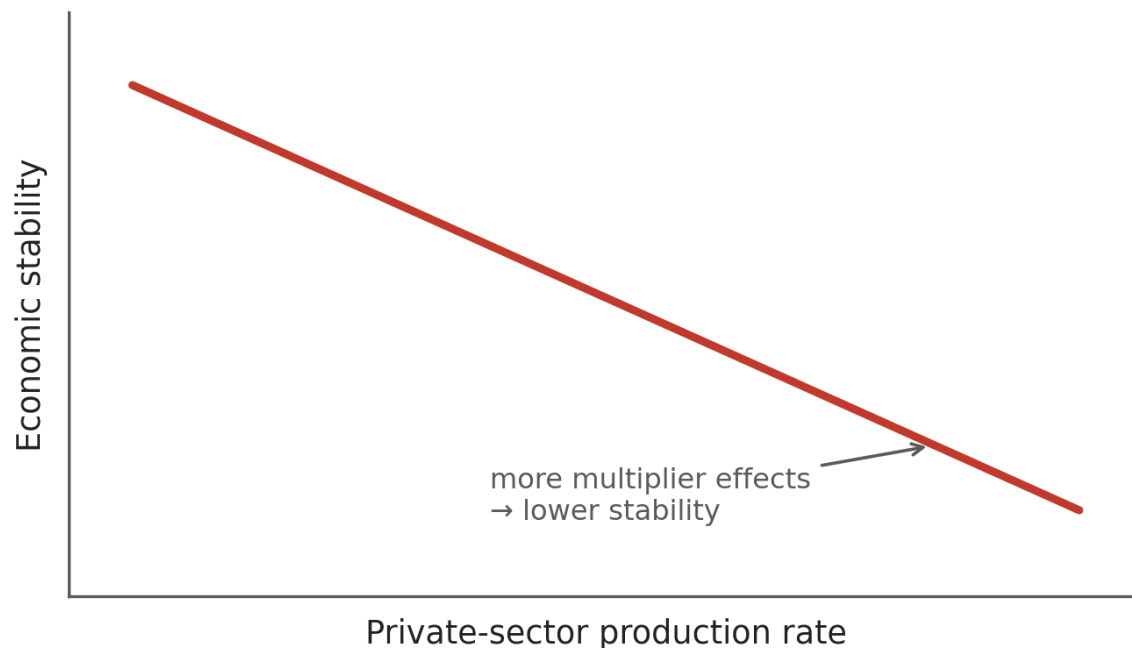


Figure 2. The same proliferation of independent decision-makers that raises growth also multiplies destabilising feedbacks, so stability declines as the private-sector share rises.

The equilibrium point **P**

Superimposing the two relationships reveals an intersection — the point **P** at which the marginal gain in growth is balanced against the marginal loss of stability (Figure 3). **P** is not a natural constant. It is the production rate at which a society has chosen to trade growth for stability, and that choice is expressed through the instruments of the public sector: legislation, fiscal policy, and monetary policy.

Growth and stability as functions of private-sector production

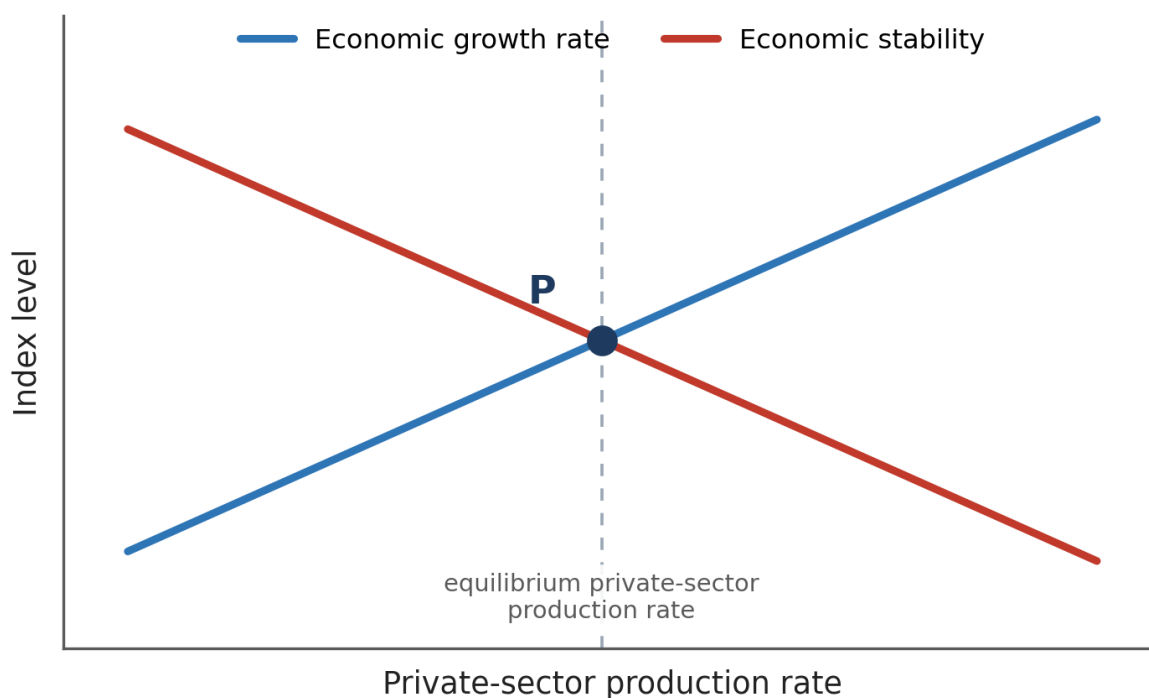


Figure 3. Point P marks the private-sector production rate at which growth and stability are in balance. Its location is set, in practice, by public-sector legislation, fiscal policy, and monetary policy.

The practical significance of P is that, although private actors enjoy the financial freedom to expand their activities, the *level* at which the private-sector production rate settles is ultimately determined by public decisions. The public sector, in this framing, is the steward of the growth–stability balance.

The erosion of control in a multipolar world

Here the diagnosis sharpens. Private production is now global, but the public decisions that are supposed to anchor P are taken by many separate institutions in a multipolar world. Each acts within its own jurisdiction, and each is subject to the same competitive logic it is meant to temper — competing for capital, for industry, and for tax base. The result is that the collective capacity to set and hold P has weakened precisely as the private sector has globalised. No single authority is positioned to manage the growth–stability balance at the scale on which the economy now actually operates.

The case for an autonomous global public sector

If the balance is set by the public sector, and if the relevant economy is global, then the logical remedy is a public-sector function that operates at global scale and is insulated from the competitive pressures that erode national policy. This paper refers to that function, following a long tradition discussed in the closing section, as a world federal

government — understood here in its narrowest economic sense: an autonomous, global decision-making body able to set and defend the growth–stability balance through coherent financial and monetary policy.

The two sections that follow develop those two policies in turn. The financial policy is redistribution; the monetary policy is the issuance of a gold-backed global currency. The central claim of the paper is that, with programmable money, the two can be unified in a single instrument.

Financial policy: redistribution as the stabilising instrument

Technological deflation and the limits of conventional monetary policy

Information technology contributes enormously to society and to the world economy, and it does so in a way that is fundamentally deflationary. Productivity gains lower the cost of supply; the same innovations lower the cost of consumption on the demand side, reducing the cost of living. Artificial intelligence and the Internet of Things are expected to deepen both effects. The economy therefore faces sustained deflationary pressure on both sides of the market at once.

Against this backdrop, a monetary policy whose primary aim is to control the inflation rate becomes progressively harder to operate, and the gap between the policy framework and the underlying market fundamentals widens. Quantitative easing, the principal unconventional tool of the past fifteen years, has had the side effect of concentrating wealth: by inflating asset prices it has rewarded existing asset-holders, lowered effective liquidity for everyone else, and widened the income gap. The consequence is no longer merely economic. The resulting social instability erodes the foundation on which a healthy economy depends.

The conclusion the paper draws is not that deflation must be resisted but that it should be accepted and shared. If technological progress lowers the cost of living, the appropriate response is a social system that distributes the resulting surplus broadly, satisfying growth and stability simultaneously. Many redistributive efforts already exist, public and private; what is missing is a clear, system-level policy that shares economic stability deliberately rather than incidentally.

A formal model of redistribution

The fundamental concept is redistribution that secures the cost of living of lower-income households. To make the idea precise, take the simplest possible redistribution schedule, a straight line:

$$y = ax + b, \quad 0 < a < 1, \quad b > 0,$$

where x is pre-distribution income, y is post-distribution income, a is the distribution (retention) rate, and b is an absolute living-standard floor. Let $f(x)$ denote the population distribution over pre-distribution income.

A self-financing redistribution is one in which the total amount transferred to those who receive more than they had equals the total amount contributed by those who receive less. Below the break-even income $x = b/(1 - a)$ households are net recipients ($y > x$); above it they are net contributors ($y < x$). The balance condition is therefore

$$\int_0^{\frac{b}{1-a}} f(x) (y - x) dx = \int_{\frac{b}{1-a}}^{\infty} f(x) (x - y) dx.$$

This is simply the continuous, schedule-based generalisation of a progressive tax-and-transfer system (Figure 4). Its virtue for the present argument is that it expresses redistribution as a rule on incomes — exactly the kind of rule that can later be written into the logic of a currency.

The simplest model of redistribution (progressive)

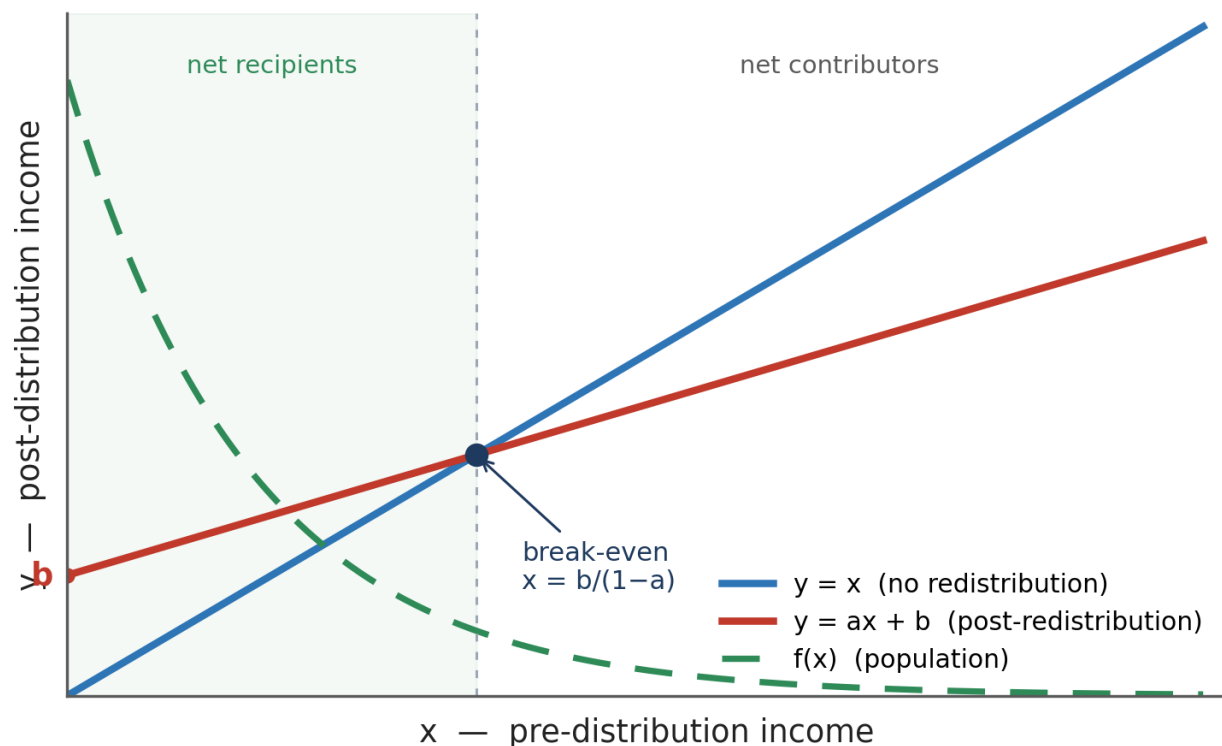


Figure 4. The simplest redistribution schedule. Households below the break-even income $b/(1 - a)$ are net recipients; those above it are net contributors. The schedule is calibrated so that contributions and receipts balance.

Universal basic income as the limiting case

The most consensual form of redistribution is the one in which everyone receives the same fixed grant. Setting the schedule to $y = x + b$ — that is, $a = 1$ — gives every household its pre-distribution income plus a uniform floor b . This is universal basic income: an unconditional payment made to every individual, on an individual basis, without means testing and without a work requirement.³

To size such a grant, the paper anchors b to the median of the income distribution — the value that divides the population into two equal halves:

$$\int_0^b f(x) dx = \int_b^\infty f(x) dx.$$

Distributing b to the entire population then implies a total redistribution

$$R = b \int_0^\infty f(x) dx,$$

which serves as the reference quantity for setting the transfer-fee rate that funds the program (Figure 5).

³ The canonical statement is Philippe Van Parijs, “The Universal Basic Income: Why Utopian Thinking Matters, and How Sociologists Can Contribute to It,” *Politics & Society* 41, no. 2 (2013): 171–182.

Universal basic income as the limiting redistribution case

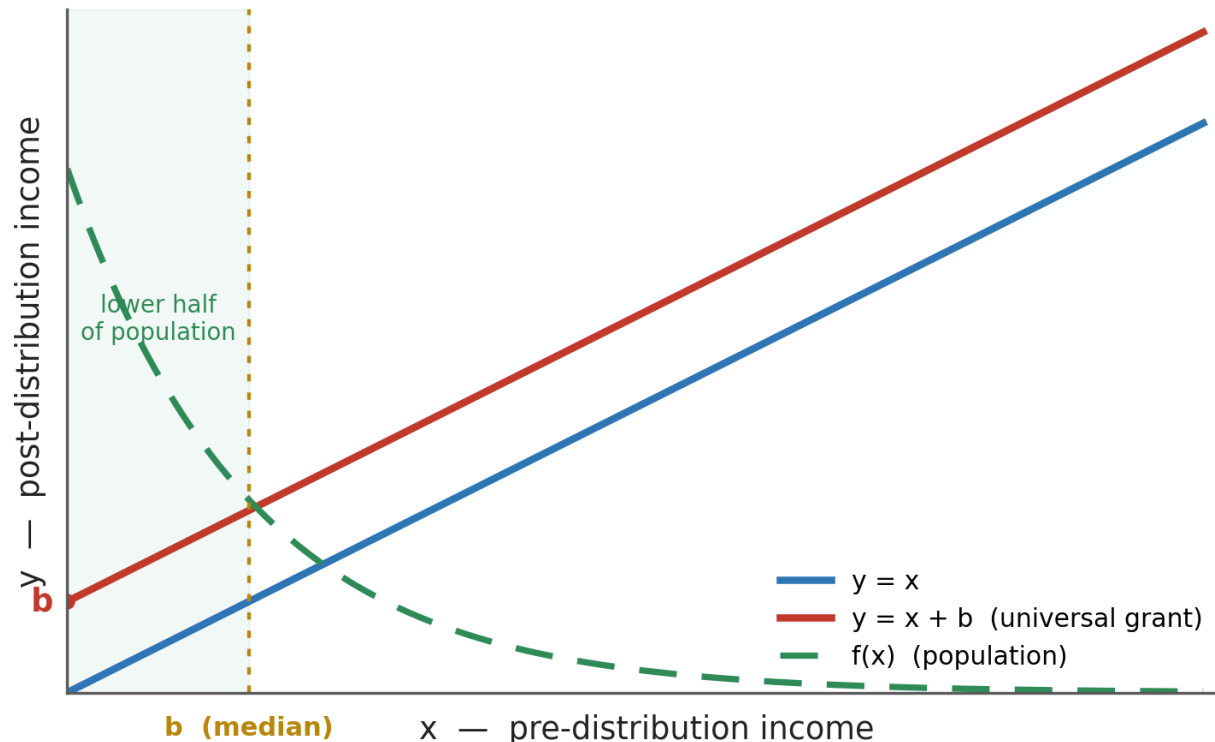


Figure 5. Universal basic income is the limiting case $a = 1$: every household receives a uniform grant b . Anchoring b to the median ensures that at least half the population are net beneficiaries.

The political economy of this choice is decisive. Because the grant is anchored at the median, at least half of all people receive more than they contribute. A measure that benefits a majority, by construction, can attract durable democratic consent — which is why universal basic income, embedded in the currency, is described here as the most consensual program available. Redistribution need not always take a monetary form: it can also be delivered as public sanitation, environmental preservation, or other services, particularly for communities whose way of life depends less on the monetary system. But the monetary grant is the simplest expression of the principle and the one most naturally embedded in money.

Monetary policy: a gold-backed global currency

Revaluation: re-anchoring trust in gold, digitally

Through the modern monetary era the U.S. dollar has served as the key global currency. As a fiat currency it is backed not by a commodity but by trust — trust engendered through legitimacy and international recognition. Yet the dollar, like the other major currencies, was itself convertible to gold until 1971, when an ounce could

be exchanged for thirty-five dollars. Gold served for centuries as a stable common measure of value, and the nature of the trust that money carries has been in flux ever since that link was severed.

This paper proposes a single global currency, generated on internet infrastructure as a blockchain for everyday use, and it proposes to re-anchor that currency to gold — not by warehousing bullion for every unit, but by **defining the unit as proportional to a weight of gold**. The illustration used throughout is the simplest one: one unit equals one gram. The purpose of the anchor is twofold. It re-establishes a stable, externally verifiable measure of value, reaffirming the relationship between the physical world and the virtual network on which the currency lives; and it provides the issuer with a macroeconomic lever, since the ratio of gold reserves to total issuance can be adjusted to manage liquidity. This re-anchoring of monetary trust to gold, expressed digitally, is the second of the three R's: **reevaluation**.

Transfer fees: redistribution without taxation

The mechanism that unites monetary and financial policy is the transfer fee. Any currency system can levy a small fee on transactions to cover the cost of issuance and operation. The proposal is to set that fee deliberately above bare operating cost and to direct the surplus to the redistribution defined in the previous section. Because millions of transfers occur every hour across a global currency, even a fee well below one percent generates a continuous, broad-based revenue stream.

The significance of funding redistribution this way is that it requires no government tax system. The redistribution rule can be stated transparently and applied uniformly; the currency itself becomes an instrument of welfare. This is the most fundamental sense in which the currency embeds public policy: a universal basic income financed automatically by the act of transacting, rather than appropriated and disbursed by a fiscal authority. The total redistribution R derived above provides the reference point against which the fee rate is calibrated.

The dual mandate of the issuer

The issuer of such a currency carries two operations that correspond to the two scales of the economy. At the microeconomic scale it administers redistribution — the daily distribution of the basic income and the collection of the fees that fund it. At the macroeconomic scale it manages gold as the principal means of asset management, adjusting the reserve ratio to maintain liquidity and stabilise value. Stabilising the new currency against gold can, in turn, help stabilise existing legal currencies as the new unit is introduced into exchange markets.

The governance of the issuer is examined in detail later in this paper. The essential principle is stated here: the issuing entity should be independent of the world government in the way that legislative and judicial functions are independent of one another, yet it should remain within the public sector. Every user of the currency should

have the right to participate in its governance — by election, by rate-based decision-making, and through direct and indirect means — so that the currency and its issuer earn trust and legitimacy by consensus.

The R-coin concept

Redistribution, Revaluation, Revolution

The three principles examined above give the working name its meaning.

Redistribution is the embedded universal basic income, financed by transfer fees rather than taxes and extended, through biometric identity, even to people outside the conventional financial system. **Revaluation** is the re-anchoring of monetary trust to gold, expressed digitally, with the unit defined as a weight of gold. **Revolution** is the deliberate departure from the decentralised-cryptocurrency model, in favour of a public, accountable institution.

Why not a decentralised cryptocurrency

Bitcoin was the first blockchain, and its design — decentralised, permissionless, governed by no one — has shaped a decade of experimentation. That experiment has also exposed the limits of the model for the role proposed here. Contested development priorities fractured the market into many competing coins; the assets have been used for illicit transactions; and their volatility has led most observers to treat them as speculative instruments rather than as currencies fit for everyday economic life.

The lesson this paper draws is not that the technology is unsuitable but that decentralisation for its own sake is the wrong objective for money that is meant to stabilise the economy, society, and daily life. The cost of maintaining a stable legal tender is real and can be measured; it is a cost worth bearing. The R-coin project would therefore begin as a private, non-profit organisation aiming to create a global currency system, with the explicit goal of evolving into a public entity — a global Central Bank Union — and preserving its public authority as it does so. This is the third R: a **revolution** in how a global currency is governed, away from anonymity and towards accountable, consensual public stewardship.

The Central Bank Union

The institutional embodiment of the proposal is the Central Bank Union (CBU): an organisation formed by the world's central banks to issue and manage the global currency and to operate the payment system on which it runs. Central banks are well suited to this role. Most are already structurally independent of their governments; they possess the operational expertise to run large-value settlement systems; and a union of them can act with the global reach the diagnosis requires while remaining within the public sector. The CBU is, in effect, the autonomous global public-sector function

identified earlier — the body capable of setting and defending the growth–stability balance through a currency that carries financial policy within it.

System architecture

The blockchain layer

The currency is modelled as a native token on a scalable, high-throughput Layer-1 blockchain. The chain must interoperate with other systems rather than stand apart from them: it must connect digitally to private payment networks, to contract systems, and to national central bank digital currencies (CBDCs). During the foundational phase, central banks issue their own CBDCs and the global unit functions as the common native token that links them.

Off-chain payments are permitted up to limited amounts, so that everyday commerce does not depend on continuous connectivity. Those limits would be waived automatically when digital infrastructure is degraded — for example by a natural disaster or a systems failure — so that the currency remains usable precisely when it is needed most.

Validators and consensus

Central banks act as the validators of the chain. Consensus is not left to anonymous mining power: each validator is an identified public institution operating under a single organisation or a shared regulatory framework. Voting rights can be assigned in one of three ways — one vote per validator; a stake proportional to gold reserves (a proof-of-stake variant); or a mixed balance of the two — with material decisions requiring the agreement of more than two-thirds of validators. Because most central banks are structurally independent of their governments, the arrangement keeps day-to-day monetary operation at arm's length from political direction while remaining within the public sector.

A supporting organisation can be established to assist the CBU: maintaining regular communication, offering advice, performing audits, and developing and maintaining the open-license resources of the system. Ideally this body takes the form of a public entity or a well-regulated decentralised autonomous organisation (DAO).

Privacy, encryption, and the two-tier model

For privacy and security, the global currency circulates initially on an encrypted network rather than as a fully public chain. The exception is the CBU's own account, whose transactions are public in order to maintain transparency over the institution that issues the money.

Privacy is reinforced by where data lives. Central banks are generally independent of their governments, so the chain need obtain only users' wallet addresses from

government databases, not their full identities. Where an account-based system struggles to retain complete payment history, index files for each wallet can be held where they are generated — at commercial banks, in personal applications, or in cloud storage — and synchronised periodically with the main chain, which strengthens privacy on a globally distributed ledger. Validators may scan only anonymous, aggregated on-chain records; analysis is performed on that bulk data using AI or related computing, and any collection of additional information requires that the affected users be notified.

This design follows the two-tier pattern now emerging in national CBDC research, in which a wholesale layer operated by central banks underlies a retail layer intermediated by banks and service providers.⁴

Access authorisation

Access to transaction data is tiered to match responsibility. The table below summarises who may see what.

Access authority	Transaction category visible
Public	All transactions of the CBU account. Published data may include certain first-layer transactions once the two-tier retail system is operational.
Central Bank Union (CBU)	Principally cross-border transactions, and all data on each central bank's cross-border activity.
Central banks and their local governments	Each domestic or alliance transaction within their own jurisdiction.

Central banks and their local governments may deploy the global currency on their own CBDC systems provided those systems comply with the CBU's universal regulations and standards, and they may delegate systems management to the CBU under a CBDC option.

Resilience and circulation control

Even when the basic income is distributed to personal wallets worldwide, domestic jurisdictions retain control over circulation as retail payment. In practice this means a person may hold the global unit in a wallet but use it only after conversion to local legal tender, so that the global currency circulates domestically at the wholesale level while national authorities continue to govern retail money. To address the digital divide,

⁴ See, for example, the People's Bank of China, *Progress of Research & Development of E-CNY in China*, and the Hong Kong Monetary Authority, *e-HKD: A Technical Perspective*. Both documents informed the access model described here.

distributable funds are held in reserve for users who register late, so that no one is excluded by the timing of their access to the system.

A note on next-generation hardware

The R-coin initiative is described by its author as a hybrid hardware-and-software project. Alongside the currency software it explores longer-horizon hardware research — logical architectures and next-generation computing concepts, including plasmon-based logic gates — intended to support the scale and security demands of a global payment system over time. This work is exploratory and is not on the critical path for the monetary proposal; it is noted here for completeness, and the supporting documents are maintained in the project repository.

The universal basic income program

Sizing the grant

To make the program concrete, consider a grant of **five units per person per year**, with one unit defined as one gram of gold. The grant is intended for the world's population aged fifteen and over — approximately 6.1 billion people — and is sized to lift recipients above the international extreme-poverty line.⁵ The implied annual budget is

$$5 \text{ g} \times 6,100,000,000 \text{ people} = 3.05 \times 10^{10} \text{ g} \\ = 30,500 \text{ tonnes of gold-equivalent per year.}$$

Two points deserve emphasis. First, the arithmetic is internally exact: five grams multiplied by 6.1 billion people is thirty thousand five hundred tonnes, and it is this 6.1-billion adult population — not a smaller figure — that the budget assumes.⁶ Second, the grant is a *floor*, not a ceiling. The amount can be raised, but increases would feed through to local economies as inflation and would interact with the global debt market,

⁵ The illustration uses the World Bank's then-current extreme-poverty line of US\$1.90 per day (2011 PPP, as of 2015). The line was subsequently updated to US\$2.15 per day (2017 PPP) in 2022; at prevailing gold prices, a grant of five grams per year remains in the neighbourhood of this threshold. Population aged 15+ is drawn from United Nations estimates (2024).

⁶ An earlier presentation of this calculation stated the population as "50 million," which is inconsistent with the thirty-thousand-five-hundred-tonne result; five grams times fifty million people is two hundred and fifty tonnes. The figure is corrected here to approximately 6.1 billion adults, which both matches the stated budget and reflects the actual global population aged fifteen and over.

so the level must be set with care. Jurisdictions that wish to provide more can fund additional domestic basic income from their own domestic transfer fees.

It is worth situating thirty thousand five hundred tonnes in context. It is a large quantity relative to the roughly two hundred thousand tonnes of gold ever mined, which is exactly why the budget is structured as a one-time gold-denominated asset issuance rather than as a requirement to take physical custody of the metal. The gold anchor is a unit of account and a stabilisation reference; it does not require the CBU to vault the entire annual budget in bullion.

Funding mechanics

Funding proceeds in two stages. To begin, the CBU — the union of central banks operating the payment system — issues the gold-asset debt corresponding to the program's budget once, and then repays it over time by collecting transfer fees. In the foundational phase, global tax contributions and the transfer fees on national CBDCs together defray the cost of the daily distribution. As the global unit gains liquidity, its own transfer fees are expected to cover the distribution costs on their own, and the fee can settle at a level expected to be below one percent initially.

The CBU sets the parameters of this system: the transfer-fee rates, the distribution amounts, the schedule of any further issuance, and related operations. Fees can be collected from wholesale payments, from retail payments, or from both, and they can be either added to a transfer or deducted from it. This flexibility lets the CBU design fee models that maximise the public interest — for example, drawing the cost of system maintenance and security from wholesale activity while funding the basic income from retail payments.

Inclusion: biometric identity and the digital divide

A global basic income is only as universal as its ability to reach people. Conventional financial infrastructure excludes large populations who lack bank accounts, smartphones, or internet access. The program therefore contemplates biometric authentication to establish a unique identity for each recipient, enabling redistribution to reach individuals the existing system cannot. Combined with the policy of holding distributable funds for late registrants, biometric identity is what allows the grant to be genuinely universal rather than universal only among the already-connected.

Universal basic income as a smart contract

Brought together, these elements describe a universal basic income implemented as a smart contract embedded in the currency. Identity verification, eligibility, the periodic grant, the collection of fees, and the recycling of those fees into the next round of distribution are all expressed as rules executed by the payment system itself. The CBU's payment infrastructure can provide the supporting operations this requires: verification of the identity of every individual holding a basic-income account, domestic

government registration, voting systems, and certification of social-service accounts, among others.

Governance and operations of the CBU

The CBU's standing operations follow from its dual mandate. They can be grouped under four headings.

Establishing the basic-income parameters. The CBU sets the grant and the rules around it, with the expectation that the program steadily corrects disparities across and within economies.

Price stability. The CBU oversees price developments at national and sub-national levels and stands ready to vary the grant where conditions require — for instance, a temporary increase to cushion a population facing rapidly rising prices.

Labour-market stability. The CBU likewise monitors labour markets at the national and local level. The basic income can be designed to support, rather than discourage, participation — for example by adding a temporary supplement for those actively searching the job market.

Security and the monitoring of illicit activity. Protection of the system and detection of illicit use are expected to be well structured and developed in cooperation with central banks while remaining independent of them, exploiting the properties of the distributed ledger described earlier.

Underpinning all four is the governance principle stated earlier: the CBU is independent in its operation but remains a public institution, and the people who use the currency hold the right to participate in its governance. Independence and accountability are not in tension here; they are the two conditions that together allow a global currency to earn lasting trust.

Implementation roadmap

The transition is staged so that capability and trust are built before scale. The foundational phase establishes national CBDCs and the underlying chain; the second phase introduces the global currency and begins distribution of the basic income; the third phase matures the CBU into a permanent public institution. The phases are milestone-gated rather than tied to fixed dates, and each depends on the demonstrated success of the one before it (Figure 6).

Sequential, milestone-gated rollout

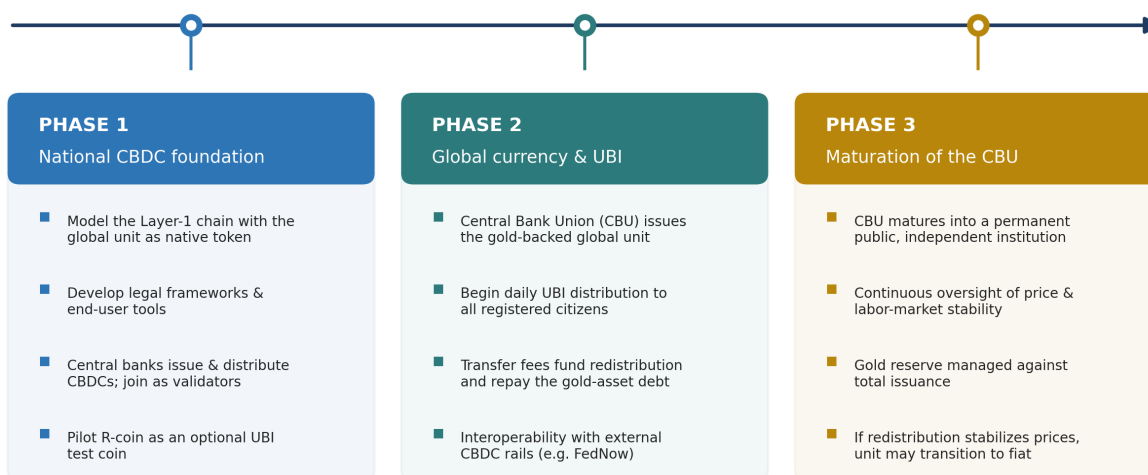


Figure 6. A milestone-gated rollout. Each phase builds the capability and the public trust required by the next; later phases proceed only as earlier ones are demonstrated to work.

In the foundational phase, the chain is modelled with the global unit as its native token, legal frameworks and end-user tools are developed, central banks issue and distribute their CBDCs and join as validators, and the system can be trialled as an optional basic-income test coin by governments that wish to plan for one. In the second phase, the CBU issues the gold-backed global unit, daily distribution of the basic income begins, transfer fees fund redistribution and repay the gold-asset debt, and the system interoperates with external CBDC rails such as instant-payment systems. In the third phase, the CBU matures into a permanent, independent public institution that continuously oversees price and labour-market stability and manages the gold reserve against total issuance. Should redistribution prove an effective price stabiliser in its own right, the currency could in time be reduced to a fiat currency — the gold anchor having served its purpose of establishing trust and stability during the transition.

Risks, safeguards, and open questions

A proposal of this scope must be candid about what could go wrong.

Local inflation and demographic effects. Where many people benefit from redistribution, the inflow can correct disparities but may also produce demographic shifts and local inflation if change arrives too quickly. The social and economic effects of the grant must be observed continuously and cautiously, and the CBU's authority to vary the grant by region exists precisely to manage this.

The scale of the gold anchor. A budget on the order of thirty thousand five hundred tonnes of gold-equivalent is large relative to global gold stocks. The proposal manages

this by treating the anchor as a unit of account and a one-time gold-denominated issuance rather than a full physical-reserve requirement, but the interaction between a gold-referenced global unit and existing gold and debt markets is a serious open question that warrants dedicated study.

Sovereignty and adoption. The arrangement asks national authorities to operate within universal CBU standards. It is designed to respect sovereignty — domestic jurisdictions retain control of retail circulation, and central banks retain their independence — but the political path to a union of central banks is the proposal's hardest practical challenge, not a technical one.

The digital divide and inclusion. Reaching people without accounts, devices, or connectivity depends on biometric identity and on holding funds for late registrants. These mechanisms must be implemented with strong privacy, consent, and data-minimisation guarantees; done poorly, the very tool meant to include people could be used to surveil them.

Privacy versus transparency. The design makes the issuer's accounts public while keeping ordinary users' transactions on an encrypted network. Maintaining that balance — transparency over the institution, privacy for the individual — is a continuing governance obligation rather than a problem solved once at launch.

Governance capture. Independence from political direction must not become independence from accountability. The participatory governance rights described in this paper, and the two-thirds threshold for material decisions, are the primary safeguards; their detailed design will determine whether they are sufficient.

None of these risks is dismissed by the proposal, and several remain genuinely open. They are set out here so that the framework can be assessed honestly by the institutions best placed to judge it.

Governance lineage: the world federal government

The economic argument of this paper — that the growth–stability balance must be managed by an autonomous public-sector function at global scale — converges with a much older institutional tradition. On 23 August 1947, in Montreux, Switzerland, world federalists from many countries gathered at the first international congress of the World Movement for World Federal Government and issued the Montreux Declaration, setting out six principles for such a government. The full text is reproduced in Appendix A.

Two of those principles bear directly on the present proposal. The sixth — the power to raise adequate revenues directly and independently of state taxes — is precisely what a transfer-fee-funded currency provides: a revenue stream that supports a global public function without a national tax apparatus. The second — the transfer to a global body of such powers as genuinely relate to world affairs — describes the limited, economically scoped authority envisaged for the CBU, which would manage the global currency while

leaving domestic monetary and fiscal sovereignty substantially intact. The CBU should be understood as a narrow, monetary realisation of the federalist idea: not a world state, but a single global public institution exercising one clearly bounded power on behalf of, and accountable to, the people who use its currency.

Conclusion

The separation of monetary policy from social policy is an artefact of an era in which money could not carry rules. Programmable digital money removes that constraint, and with it the necessity of treating the issuance of money and the sharing of prosperity as different problems solved by different institutions.

This paper has argued that the modern economy faces a structural tension between growth and stability that no national authority can now manage alone; that redistribution, in its most consensual form of a universal basic income, is the instrument capable of holding that balance; and that a gold-anchored global currency, issued by a union of central banks and funded by a small fee on every transaction, can deliver that redistribution automatically, universally, and without taxation. The result is a currency that is also a welfare instrument — *redistribution, revaluation, and revolution* combined in a single design.

The proposal is offered as a framework for deliberation. Its parameters are illustrative, its risks are real, and its hardest obstacles are political rather than technical. But its central claim is, the author submits, robust: the architecture of the next global currency is a choice, and it can be chosen so that the gains of technological progress are shared by everyone from the outset. The invitation of this paper is to treat that choice as a question worth taking seriously now, while the architecture is still being designed.

Addendum

In the process of creating this document, the author used Anthropic's Claude (paid tier) to refine and complete it.

Appendix A — The Montreux Declaration: six principles

At the first international congress of the World Movement for World Federal Government, held in Montreux, Switzerland, on 23 August 1947, delegates called upon the peoples of the world to join their work and published a declaration of six principles for a world federal government:

1. **Universal membership.** The world federal government must be open to all peoples and nations.
2. **Limitation of national sovereignty,** and the transfer to the world federal government of such legislative, executive, and judicial powers as relate to world affairs.
3. **Enforcement of world law directly on the individual,** whoever and wherever he may be, within the jurisdiction of the world federal government: guarantee of the rights of man and suppression of all attempts against the security of the federation.
4. **Creation of supranational armed forces** capable of guaranteeing the security of the world federal government and of its member states, with disarmament of member nations to the level of their internal-policing requirements.
5. **Ownership and control by the world federal government** of atomic development and of other scientific discoveries capable of mass destruction.
6. **Power to raise adequate revenues directly** and independently of state taxes.

The proposal in this paper engages chiefly with the second and sixth principles, in the narrow economic and monetary domain described above.

Appendix B — Mathematical summary of the redistribution model

Notation. x : pre-distribution income. y : post-distribution income. a : distribution (retention) rate, $0 < a < 1$. b : absolute living-standard floor. $f(x)$: population distribution over pre-distribution income.

Redistribution schedule.

$$y = ax + b.$$

Break-even income (where $y = x$):

$$x^* = \frac{b}{1-a}.$$

Self-financing balance condition (receipts below break-even equal contributions above it):

$$\int_0^{\frac{b}{1-a}} f(x) (y - x) dx = \int_{\frac{b}{1-a}}^{\infty} f(x) (x - y) dx.$$

Universal-basic-income limit ($a = 1$): every household receives a uniform grant,

$$y = x + b.$$

Median anchor for the grant b :

$$\int_0^b f(x) dx = \int_b^\infty f(x) dx.$$

Total redistribution:

$$R = b \int_0^\infty f(x) dx.$$

Worked budget (illustrative): grant of 5 g of gold-equivalent per person per year for a population of 6.1×10^9 adults gives

$$R = 5 \text{ g} \times 6.1 \times 10^9 = 3.05 \times 10^{10} \text{ g} = 30,500 \text{ tonnes of gold-equivalent per year.}$$

Appendix C — Glossary

CBDC (central bank digital currency). A digital form of a nation's currency, issued and backed by its central bank.

CBU (Central Bank Union). The proposed institution, formed by the world's central banks, that issues and governs the global currency and operates its payment system.

DAO (decentralised autonomous organisation). An organisation governed by rules encoded on a blockchain; proposed here, in well-regulated form, as a possible structure for the body that supports the CBU.

Fiat currency. Money issued by a government as legal tender, backed by trust and legitimacy rather than by a commodity.

Layer-1 blockchain. A base-layer blockchain network that settles its own transactions, as opposed to a layer built on top of another chain.

Transfer fee. A small charge on each transaction; in this proposal, set above operating cost so that the surplus funds the universal basic income.

Two-tier system. A payment architecture with a wholesale layer operated by central banks beneath a retail layer intermediated by banks and service providers.

UBI (universal basic income). An unconditional payment made to every individual, on an individual basis, without means testing and without a work requirement.